MIL-B-23394B(OS)
Amendment 5
20 September 1971
Superseding
Amendment 4
21 January 1970

MILITARY SPECIFICATION

BATTERY MARK 46 MOD 1

PRIMARY TYPE, SILVER-ZINC-ALKALI

This amendment forms a part of Military Specification MIL-B-23394B(OS) of 6 June 1966 and has been approved by the Naval Ordnance Systems Command, Department of the Navy

By this change, basic specification MIL-B-23394B(WP) is changed to MIL-B-23394B(OS).

Wherever it appears in specification delete "BUWEPS" and substitute "NAVORD".

Page 1, paragraph 2.1, under MILITARY: Delete "MIL-W-76" and its title and substitute "MIL-C-915 Cable, Electrical, Special Purpose, General Specification for".

Page 4, paragraph 3.2.3.2: Delete "shall conform to Specification MIL-W-76 or Specification MIL-W-16878" and substitute "shall conform to MIL-C-915, Type TRXF-105, or to MIL-W-16878."

Page 4, paragraph 3.2.6.1, following "(f) SERIAL NUMBER". Add: "(g) NAVAL ORDNANCE SYSTEMS COMMAND or NAVORDSYSCOM". "(h) TYPE 1 - SHELF LIFE ITEM _____". A five year expiration date, from manufacturing date, to be filled in the blank by the vendor.

Page 7, paragraph 3.4.3. Delete and substitute:
"3.4.3 Active material. The total silver content of all the active material, such as argentic oxide, argentous oxide, and powdered silver, in all cathodes of the propulsion and auxiliary cells shall be 560 +3% -0% troy ounces of silver. The total silver content of the structural components in the entire battery, such as anode and cathode grids, plate straps, terminals, and intercell connectors, shall be 140 +3% -0% troy ounces of silver. The minimum total silver requirement for active material, 560 troy ounces, and for structural components, 140 troy ounces, shall be met independently. Any proposed excess in either of these requirements shall not be permitted to reduce the minimum requirements of the other. Only silver shall be used for structural material within the envelope of the battery cells."

FSC 6135

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Page 7, paragraph 3.4.3.1: Delete entirely.

Page 7, paragraph 3.4.4: Change "sealed chamber" to "hermetically sealed chamber".

Page 7, paragraph 3.4.4.3. Delete and substitute:
"3.4.4.3 Electrolyte release mechanism. The system used to release the electrolyte from its storage container shall be either one which employs a contact type mechanism which punctures the seal on the container, or one which employs a pressure buildup within the electrolyte container to mechanically puncture or rupture the seal."

Page 8, paragraph 3.4.8, Line 4: Delete "75 plus or minus 15 degrees" and substitute "90 plus or minus 10 degrees".

Page 8, paragraph 3.4.14. Delete entirely.

Page 9, paragraph 4.1. Delete and substitute:
"4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements."

Page 10, paragraph 4.3.1.3. Delete and subtitute:
"4.3.1.3 Production sample cells. Ten cells representing propulsion and eight cells representing auxiliary sections shall be randomly selected from cells manufactured during the period from the beginning of the contract to completion of delivery of 75 percent of contract quantity of batteries. These cells shall be analyzed for silver content (3.4.2 and 3.4.3)."

Page 10, paragraph 4.3.1.5. Delete entirely.

Page 13, paragraph 4.4.4.6.1: Delete entirely.

Page 13, paragraph 4.4.5.1: Delete entirely.

Page 14, paragraph 4.4.7. Delete and substitute:
"4.4.7 Packaging and packing. The packaged battery shall be examined to ensure that it complies fully with the requirements of Section 5."

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Page 15: Replace Figure 1 with new Figure 1, attached.

Page 16: Replace Figure 2 with new Figure 2, attached.

Page 17. Add the following new paragraphs:

"4.5.2 Actuation power. A source of actuation power, as specified in 3.4.9, shall be applied to the actuating terminals of the battery. The actuating voltage shall be measured and plotted against time. Battery actuation shall be completed within 50 ms or less."

4.5.3 Angle of operation. There shall be no leakage of electrolyte during the first 9 minutes of normal discharge of a series-connected battery, or the first 30 minutes of normal discharge of a parallel-connected battery, when the battery is tilted at any angle between 0 and 45 degrees from an upright position."

Page 18, paragraphs 5.2, 5.2.1, and 5.2.2: Delete entirely.

Page 19. Add the following new paragraphs:

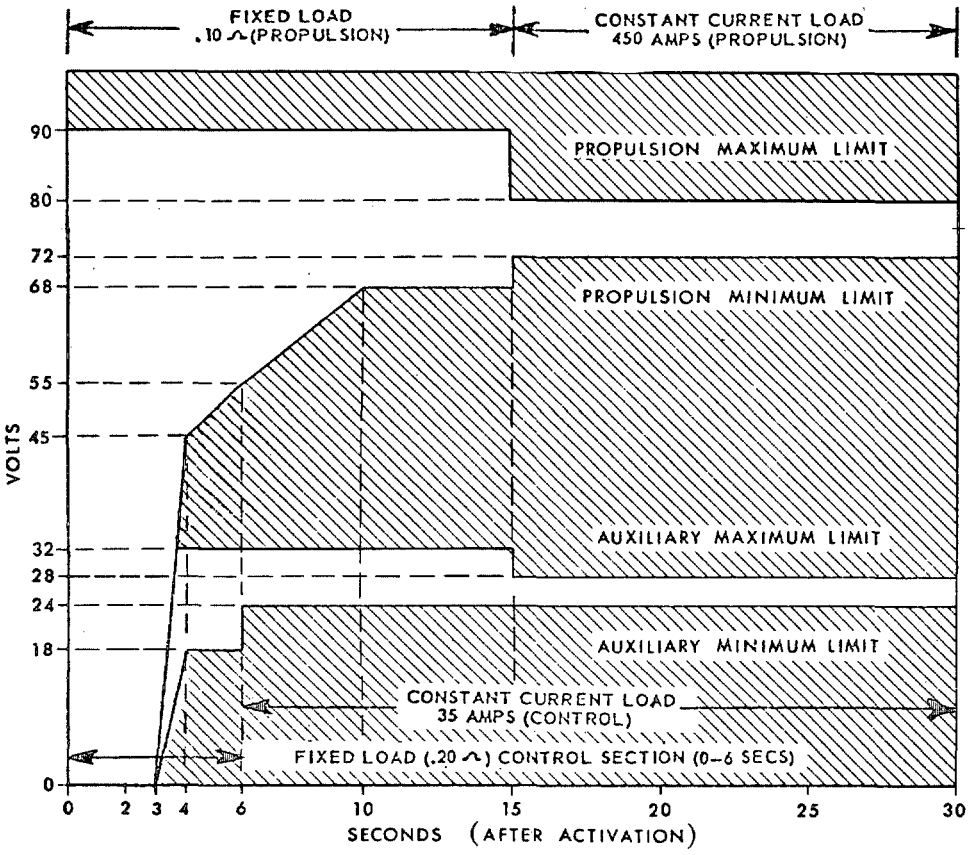
"6.4 Storage life. This battery is intended to withstand the service conditions specified in 3.3.4 and 3.3.5 for a 5-year period and be capable of providing performance characteristics in accordance with Figure 1 for a minimum of 9.5 minutes."

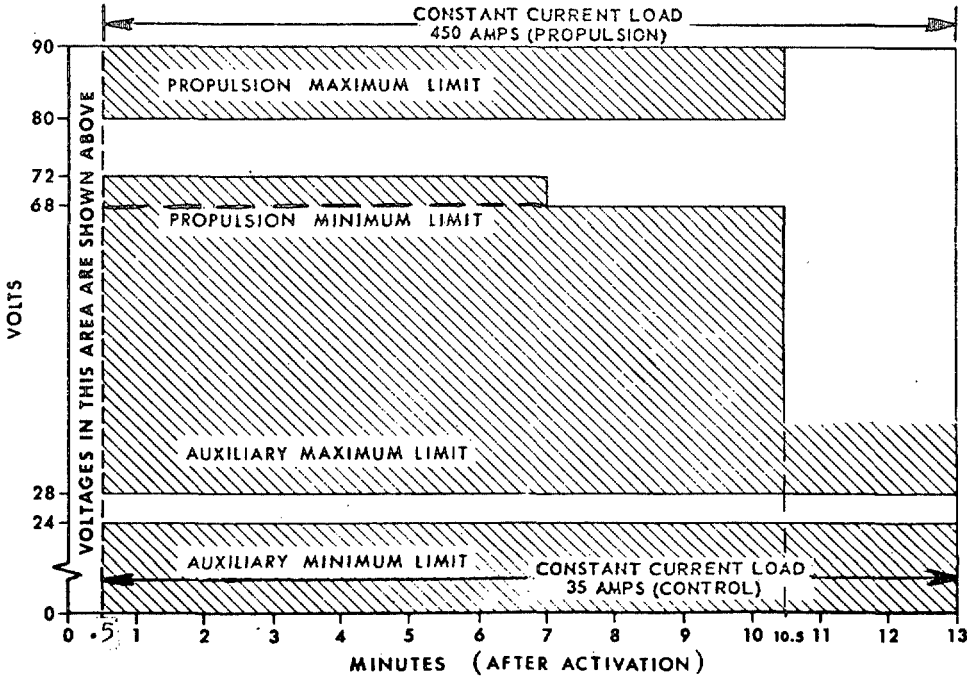
"6.5 Wet stand life. From a safety standpoint, it is highly desirable that this battery have, without benefit of an external electrical load, a wet stand life of eight or more hours following activation."

Preparing Activity
Navy - OS
Project No. 6135-N121

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SERIES CONNECTED
PRIMARY BATTERY DISCHARGE CURVES (MK 46 MOD 1)



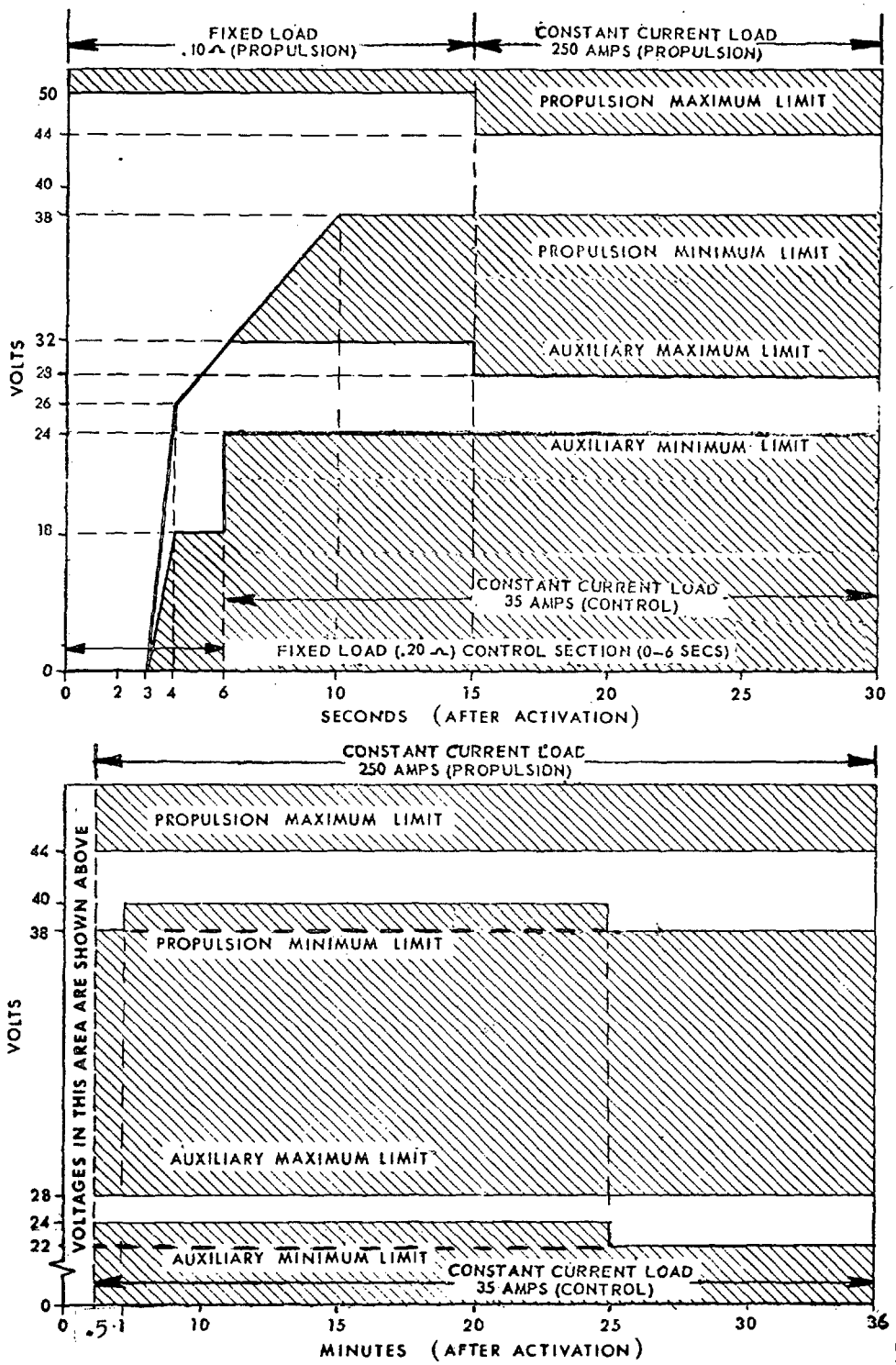


BATTERY MARK 46 MOD 1

FIGURE 1

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PARALLEL CONNECTED
PRIMARY BATTERY DISCHARGE CURVES



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FIGURE 2

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